

Efficient Quantum Fourier Transforms For Non-Group Algebras

Ben Foxman

(Yale, IBM)



Barak Nehoran

(Columbia)



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(Yale)



Overview

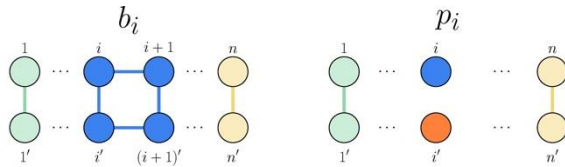
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- Specifically, QFTs **beyond groups**.

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The Fourier Transform
and **Diagram Algebras**

$$\text{FT}_G U_g \text{FT}_G^\dagger = \begin{pmatrix} \rho_1(g) & & & \\ & \rho_2(g) & & \\ & & \rho_2(g) & \\ & & & \ddots \\ & & & & \rho_n(g) \end{pmatrix}$$

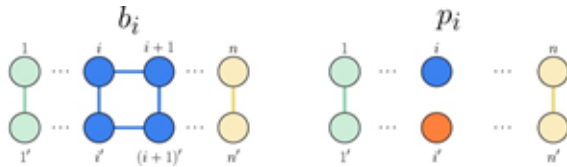


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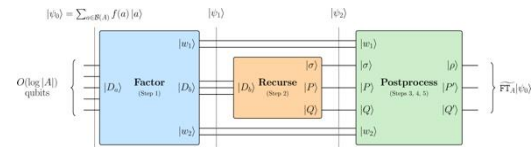
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Upgrading the “*Separation of Variables Approach*”
[DR 90, MRR 03, MRW 16]



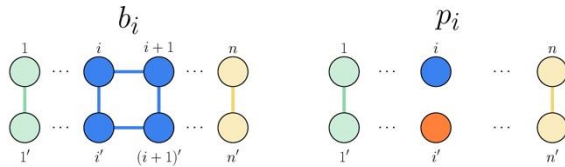
- Non-Unique Factorization
- “**Generalized**” Embeddings
- Replacing Dual Elements
- and **more...**

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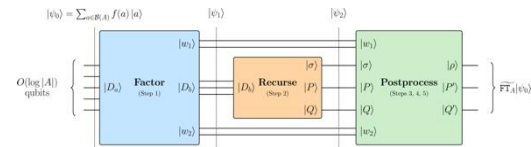
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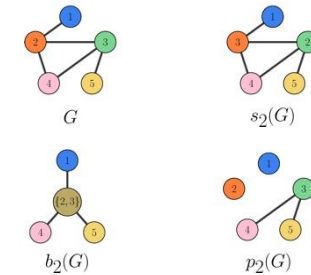


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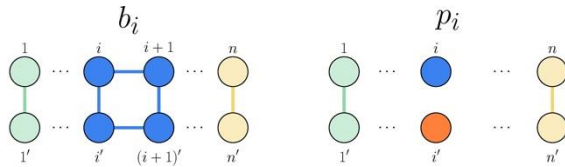


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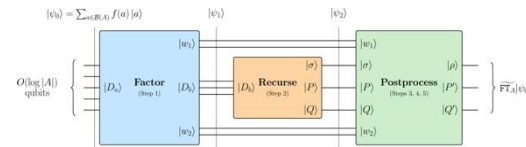
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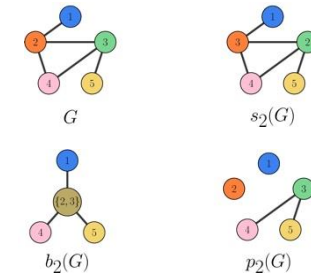


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Generalized Schur Transforms

[BCH 05, Krovi 19]

$$[U(d)^{\otimes n}, \mathbb{C}[S_n]] = 0$$

$$[O(d)^{\otimes n}, B_n(d)] = 0$$

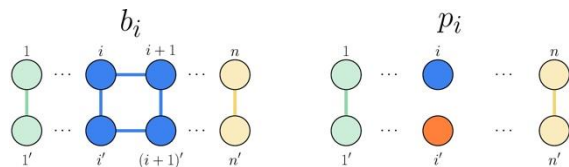
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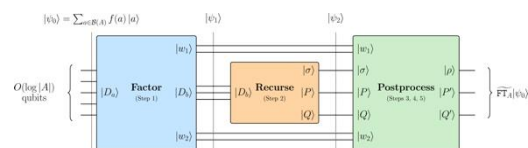
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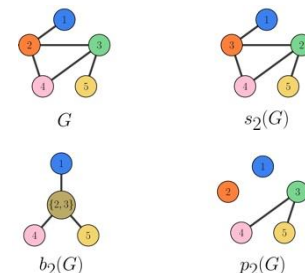


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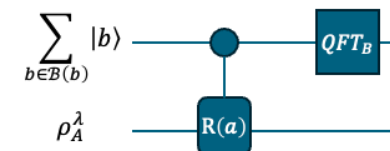
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Diagram Irrep Multiplicities
[LH 24, CHPPW 26]

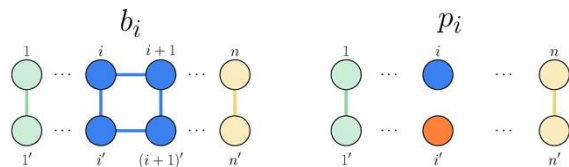


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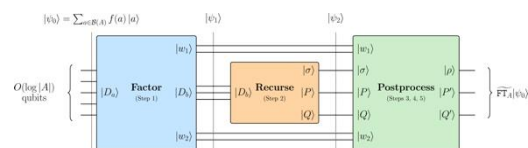
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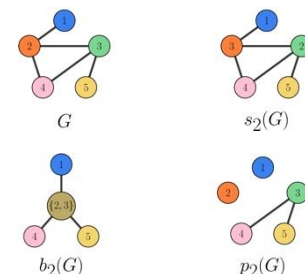


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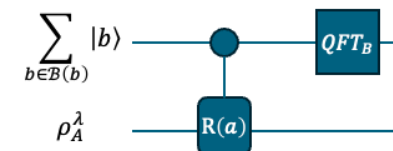
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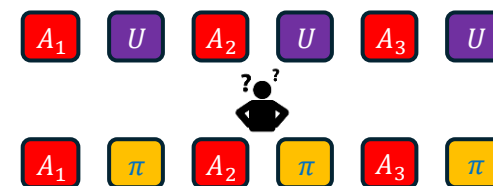
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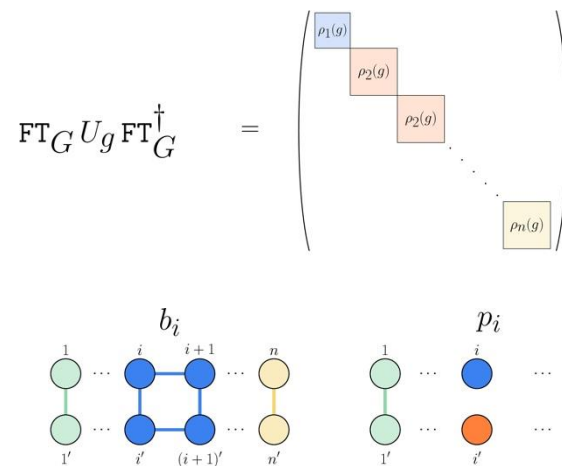
Pseudorandom Groups
[FLMNW, Upcoming]



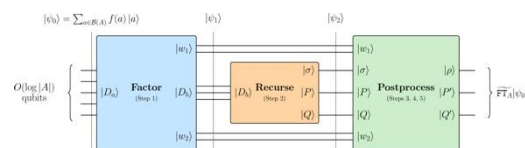
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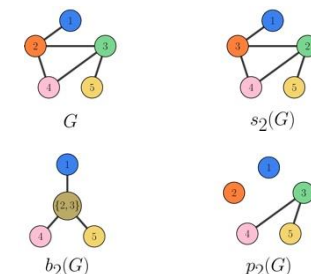


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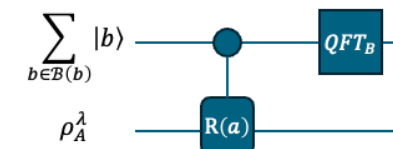
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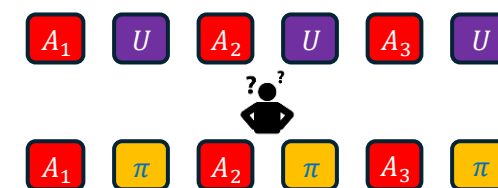
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And along the way, several new facts about the Fourier bases of *semisimple* algebras!

The Fourier Transform

- Super common task:
block-diagonalize a
set of matrices.

$$U_i = F \left(\begin{array}{cccc} \boxed{\lambda_1(i)} & & & \\ & \boxed{\lambda_1(i)} & & \\ & & 0 & \\ & & & 0 \\ & & \boxed{\lambda_2(i)} & \\ & & & \boxed{\lambda_2(i)} \\ & & & & \boxed{\lambda_3(i)} \\ & & & & & \boxed{\lambda_4(i)} \end{array} \right) F^\dagger$$

The Fourier Transform

- Super common task:
block-diagonalize a
symmetry, or
equivalently a **group**
action.

$$U_g = F$$

$$\begin{pmatrix} \boxed{\lambda_1(g)} & & & & \\ & \boxed{\lambda_1(g)} & & & \\ & & \boxed{\lambda_2(g)} & & \\ & & & \boxed{\lambda_2(g)} & \\ 0 & & & & \boxed{\lambda_3(g)} \\ & & & & & \boxed{\lambda_4(g)} \end{pmatrix} F^\dagger$$

The Fourier Transform

- Super common task: block-diagonalize a **symmetry**, or equivalently a **group action**.
 - Ex: $G \subset P_n$ is a stabilizer group

$$P = C$$

$$\lambda_j(P) = (-1)^{f(P)+z(P)\cdot j}$$
$$C^\dagger$$

The Fourier Transform

- Super common task:
block-diagonalize a
symmetry, or
equivalently a **group**
action.

- Ex: $G = \mathbb{Z}_d$.

$$U_x = \text{QFT}_d$$

$$U_x |y\rangle = |(x + y) \bmod d\rangle$$

$$\lambda_j(x) = e^{2\pi i j x / d}$$

$\text{QFT}_d^\dagger$

The Fourier Transform

- Super common task: block-diagonalize a **symmetry**, or equivalently a **group action**.

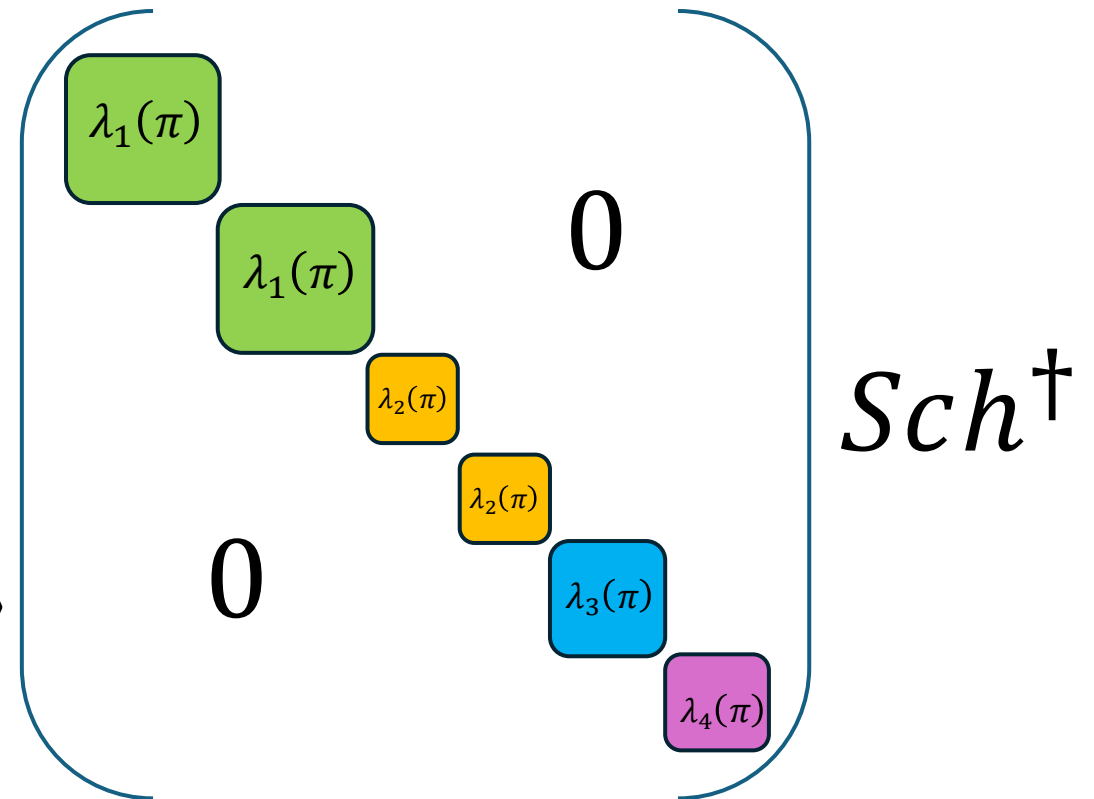
• Ex: $G = S_n$.

[Sch 27, Wey 39]

$$U_\pi = Sch$$

$$U_\pi |x_1, x_2, \dots, x_n\rangle = |\pi^{-1}(x_1), \pi^{-1}(x_2), \dots, \pi^{-1}(x_n)\rangle$$

λ_j is a *symmetry type* (symmetric, antisymmetric, etc.)



The Fourier Transform

- Super common task: block-diagonalize a **symmetry**, or equivalently a **group action**.

$$U_g = QFT_G$$

$$U_g |h\rangle = |gh\rangle$$

- This is the (left) *regular representation*, acting on $\mathbb{C}[G]$.

λ_j is a *symmetry type* (“irrep”) of G

$$\begin{pmatrix} \lambda_1(g) & & & & \\ & \lambda_1(g) & & & \\ & & \lambda_2(g) & & \\ & & & \lambda_2(g) & \\ & 0 & & & \lambda_3(g) \\ & & & & & \lambda_4(g) \end{pmatrix} QFT_G^\dagger$$

The Fourier Transform

- What happens when we generalize this to an algebra?

$$U_g = QFT_G$$

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The Fourier Transform

- Super common task: block-diagonalize a **symmetry**, or equivalently an **algebra action**.

Not unitary! 😱

$$U_a = QFT_A$$

Not invertible!



$$U_a|b\rangle = |ab\rangle$$

- This is the (left) *regular representation*, acting on $\mathbb{C}[\mathcal{B}(A)]$.

λ_j is a *symmetry type* (“irrep”) of A

$$\begin{pmatrix} \lambda_1(a) & & & & \\ & \lambda_1(a) & & & \\ & & \lambda_2(a) & & \\ & & & \lambda_2(a) & \\ 0 & & & & \lambda_3(a) \\ & & & & & \lambda_4(a) \end{pmatrix} QFT_A^\dagger$$

The Fourier Transform

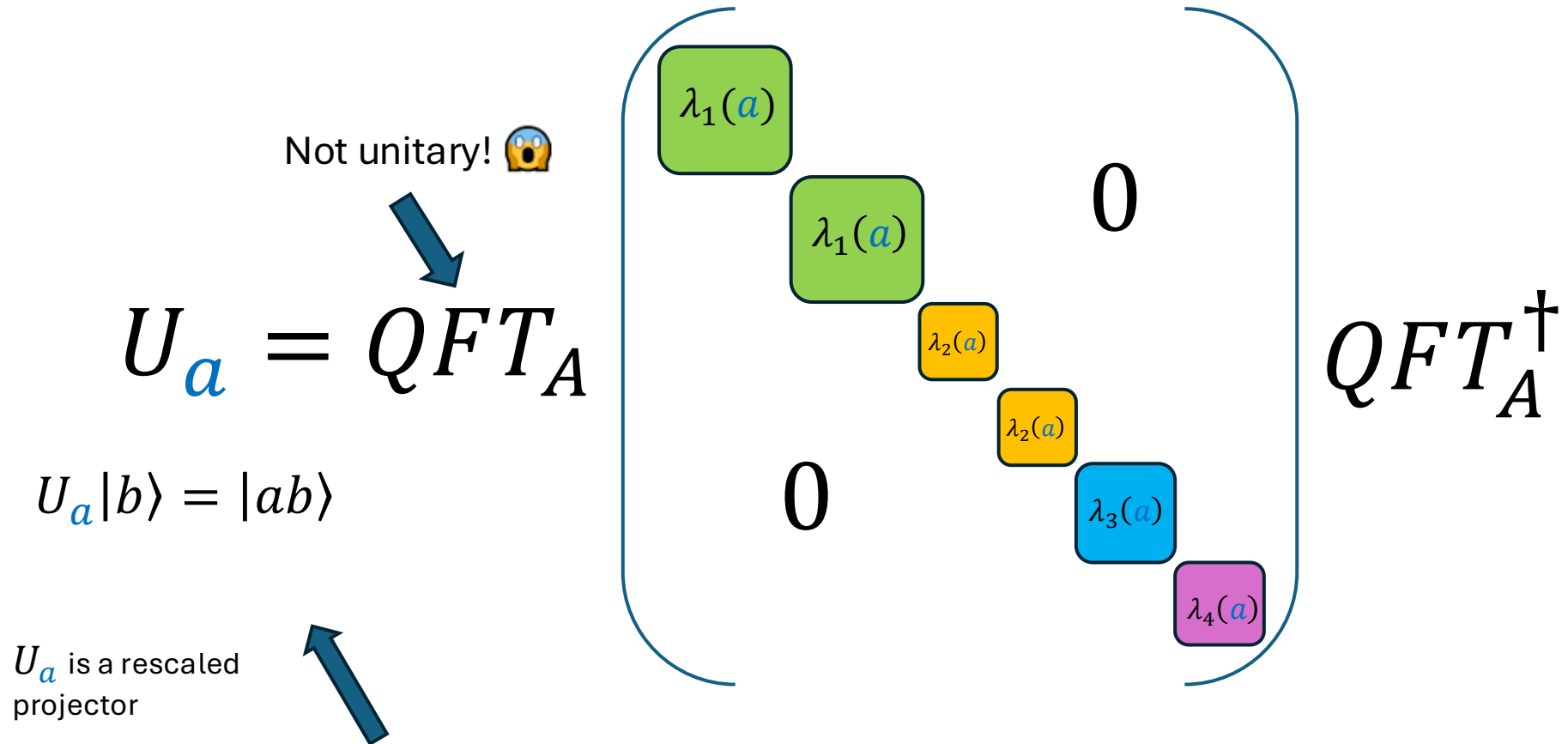
Not unitary! 😱

$U_a = QFT_A$
 $U_a |b\rangle = |ab\rangle$

$QFT_A^\dagger$

If our goal is to (efficiently) implement QFT_A on a quantum computer, then we seem to be out of luck...

The Fourier Transform

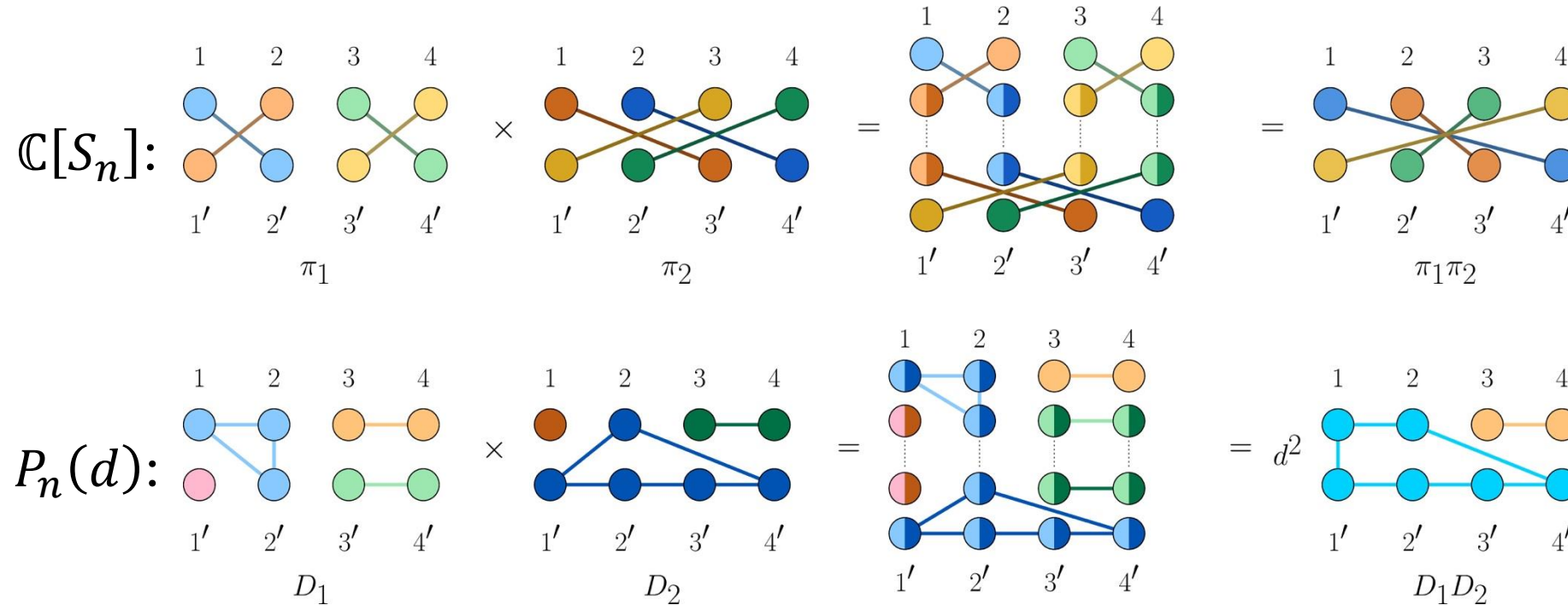


But, for some “highly non-invertible algebras”, QFT_A is still very close to being unitary!

And, with several new ideas, we can generalize (efficient) algorithmic techniques used for QFT_G .

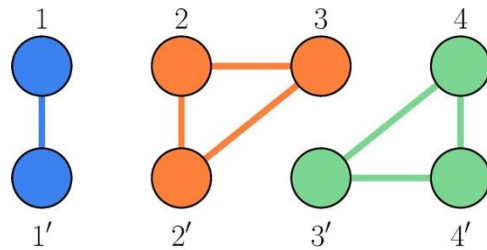
Some Non-Group Algebras

- In this work, we give efficient QFTs for several [diagram algebras](#).
- These algebras naturally generalize the symmetric group (algebra).

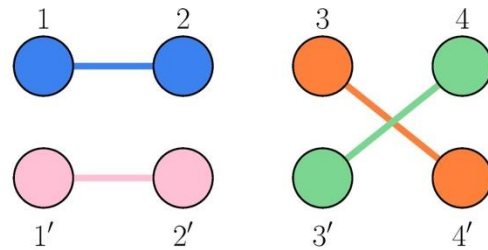


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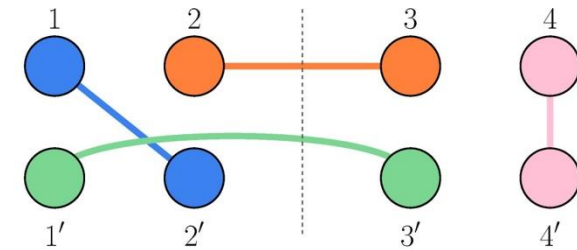
- Partition Algebra $P_n(d)$ [Mar 96]: All Diagrams Allowed
- Brauer Algebra $B_n(d)$ [Bra 37]: Matchings Allowed
- Walled Brauer Algebra $B_{r,s}(d)$ [Koi 89, Tur 89]: “Walled” Matchings Allowed



$D \in P_4(d)$



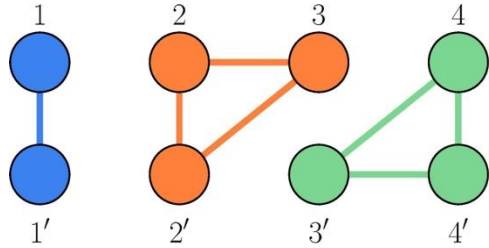
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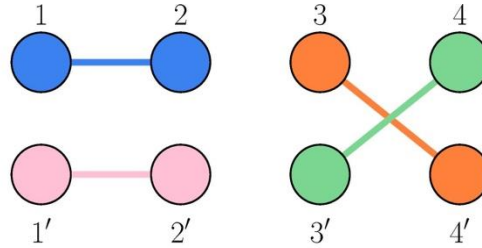
$D \in B_{2,2}(d)$

Like permutations, $|D\rangle$ can be encoded with $O(n \log n)$ qubits.

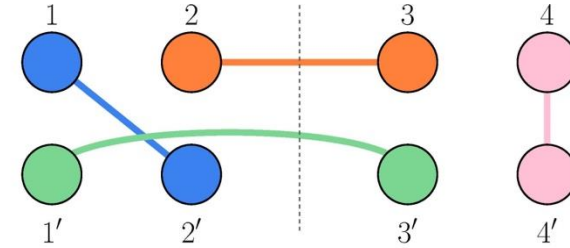
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$$D \in P_4(d)$$



$$D \in B_4(d)$$



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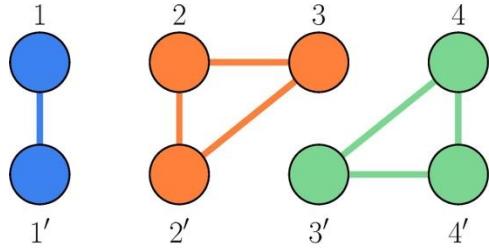
In this work, we focus on block-diagonalizing the *regular representation* $U_D|E\rangle = |DE\rangle$, implementing

$$U_D = QFT_A \begin{pmatrix} \boxed{\lambda_1(D)} & & & \\ & \boxed{\lambda_1(D)} & & 0 \\ & & \boxed{\lambda_2(D)} & \\ 0 & & & \boxed{\lambda_3(D)} \\ & & & & \boxed{\lambda_4(D)} \end{pmatrix} QFT_A^\dagger$$

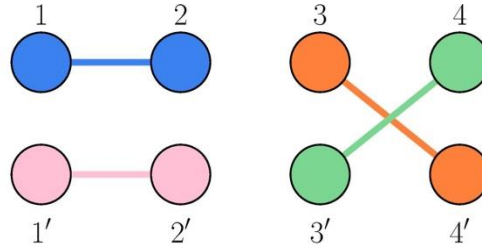
$$\begin{aligned} QFT_A|D\rangle &= \sum_{\lambda} \sum_{i,j=1}^{d_{\lambda}} \langle E_{ij}^{\lambda}, D \rangle_0 |E_{ij}^{\lambda}\rangle \\ &= \sum_{\lambda} \sum_{i,j=1}^{d_{\lambda}} \langle E_{ij}^{\lambda}, D \rangle_0 |\lambda, i, j\rangle \end{aligned}$$

But, another representation, generalizing the permutation action on n registers, is useful to have in mind.

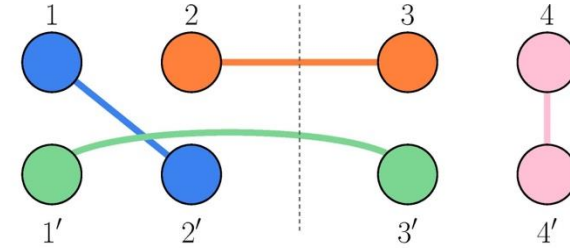
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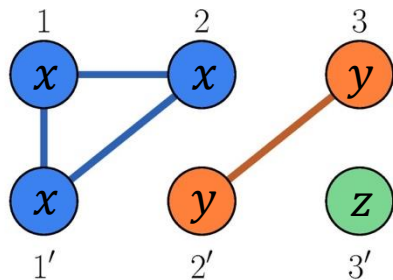


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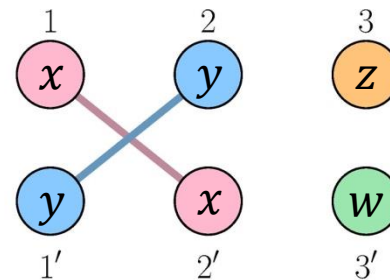


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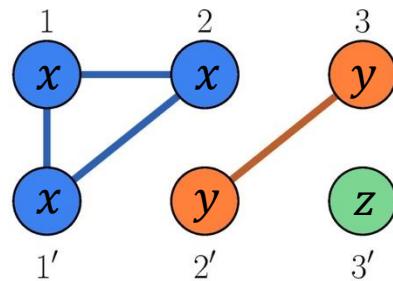
$$\mathbf{S}(b_1 s_2 p_3) = \sum_{x,y,z} |xxy\rangle\langle xyz|$$



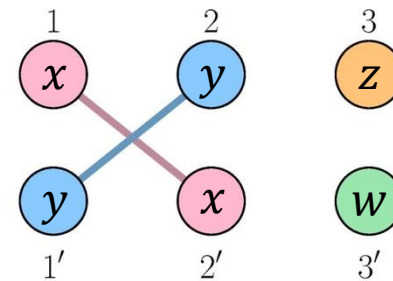
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- We show that this *Schur representation* induces an **approximately** orthogonal inner product on diagrams $|D\rangle$.
- The Schur representation induces an **exactly** orthogonal inner product for the Fourier states $|E_{ij}^\lambda\rangle = |\lambda, i, j\rangle$.
- Passing to the usual inner product, we conclude that

QFT_A is $\exp(n \log n) / \sqrt{d}$ - close to unitary.

So, we need d to be exponentially large (this is tight). However, there's an “almost-diagram” basis with no error.

The QFT for Diagram Algebras

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In the rest of the talk: an approximating unitary is *efficient* to implement.

Hidden Subalgebras

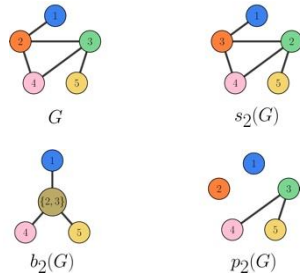
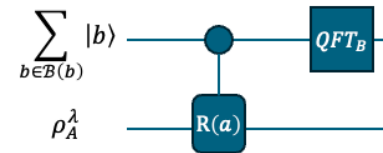


Diagram Irrep Multiplicities
[LH 24, CHPPW 26]



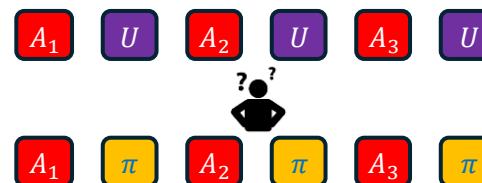
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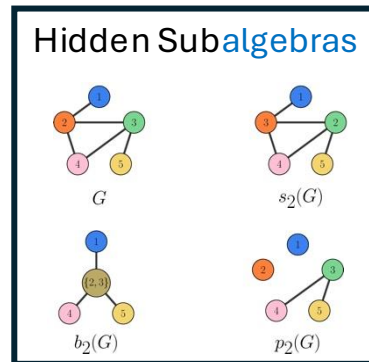
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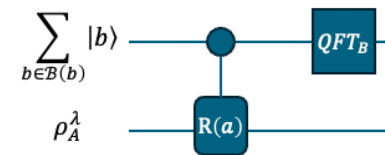
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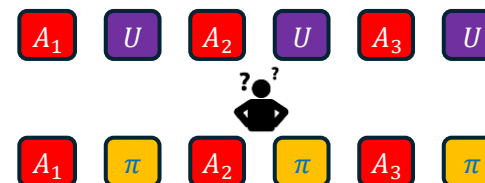
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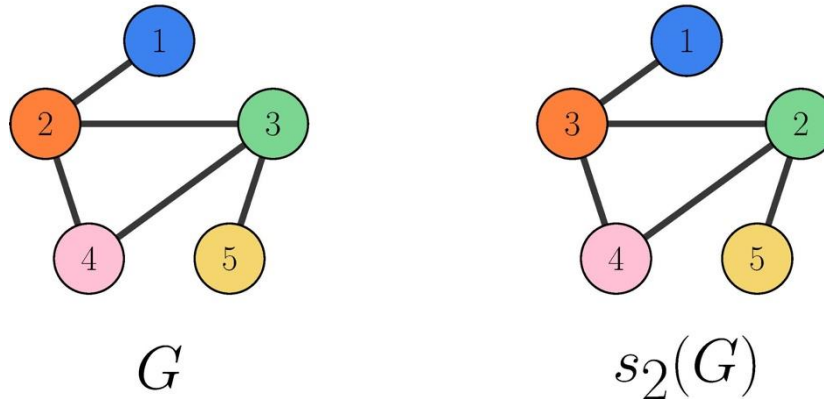


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[FLMNW, Upcoming]



The Hidden Subgroup Problem

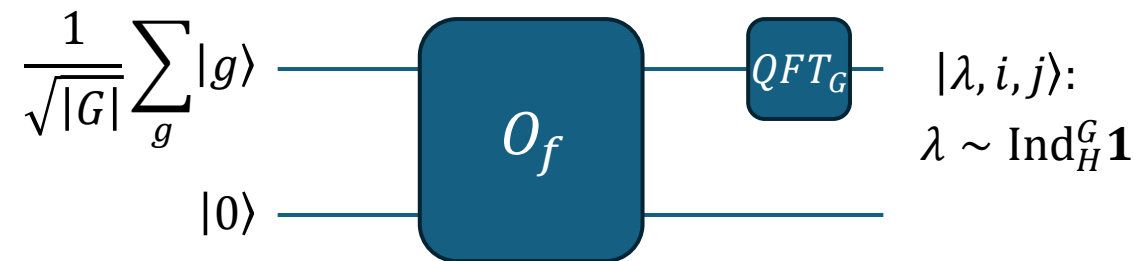
- Given oracle access to $f : G \rightarrow \{0,1\}^*$, find a subgroup H .
 - f is promised to satisfy $f(g) = f(g \cdot h)$, for all $h \in H$.
- Hidden Shift (Simon), Period Finding/Discrete Log (Shor).
- Graph Automorphism: $G = S_n$, $H = \text{Aut}(G)$.



- Graph *Isomorphism* reduces to Graph Automorphism.

The Hidden Subgroup Problem

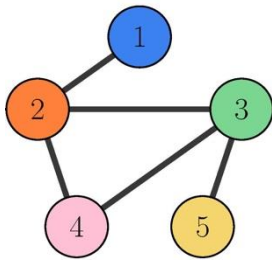
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 - f is promised to satisfy $f(g) = f(g \cdot h)$, for all $h \in H$.
- Hidden Shift (Simon), Period Finding/Discrete Log (Shor)
- Typical Strategy: **Fourier Sampling**



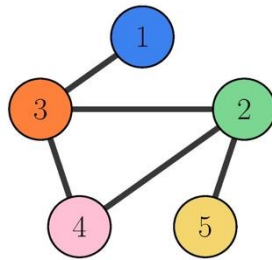
- For some non-Abelian groups (like S_n), this isn't enough [MRS 18].
- However, not much is known beyond Fourier-analytic techniques.

The Hidden Subalgebra Problem

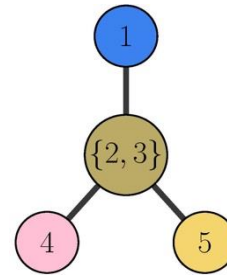
- Given oracle access to $f : A \rightarrow \{0,1\}^*$, find a subalgebra B .
 - f is promised to satisfy $f(a) = f(a \cdot b)$, for all $b \in B$.
- Returning to Graph Isomorphism...



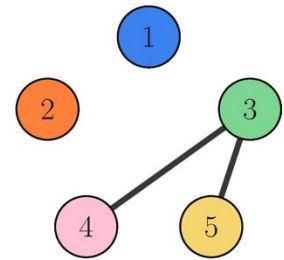
G



$s_2(G)$

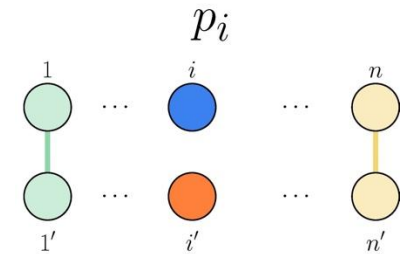
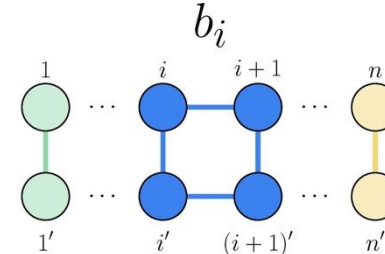
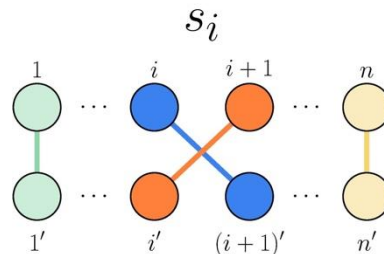


$b_2(G)$



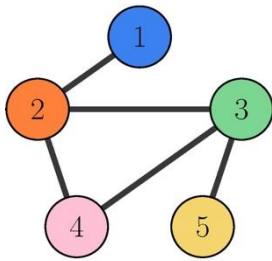
$p_2(G)$

Determining if G_1 reduces to G_2 under vertex contractions is NP-complete! [BV 87]!

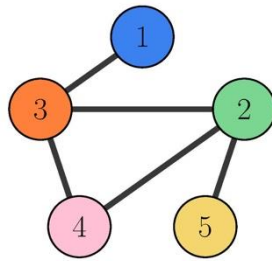


The Hidden Subalgebra Problem

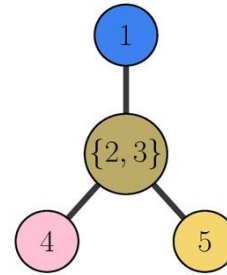
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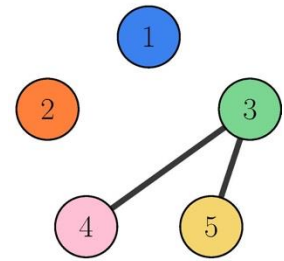
G



$s_2(G)$

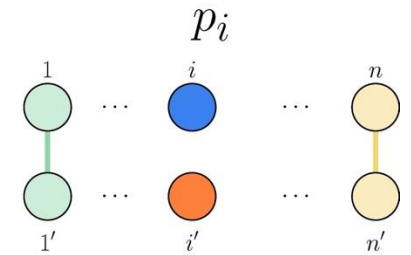
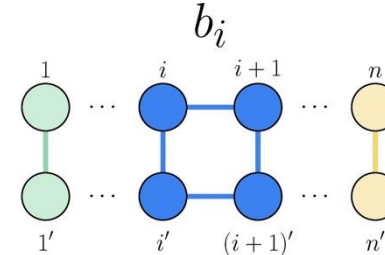
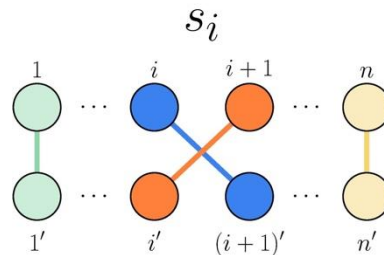


$b_2(G)$



$p_2(G)$

Conjecture: $HSAP(P_n(d))$
is NP-complete.



Applications

Hidden Subalgebras

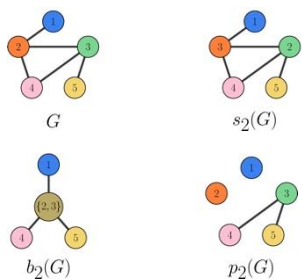
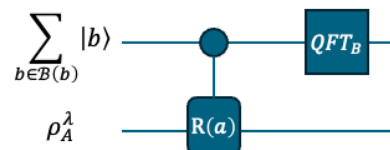


Diagram Irrep Multiplicities [LH 24, CHPPW 26]



Generalized Schur Transforms

[BCH 05, Krovi 19]

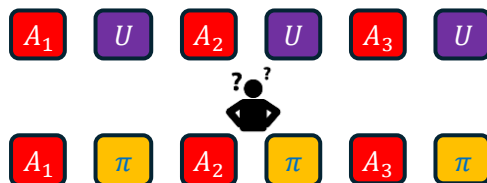
$$[U(d)^{\otimes n}, \mathbb{C}[S_n]] = 0$$

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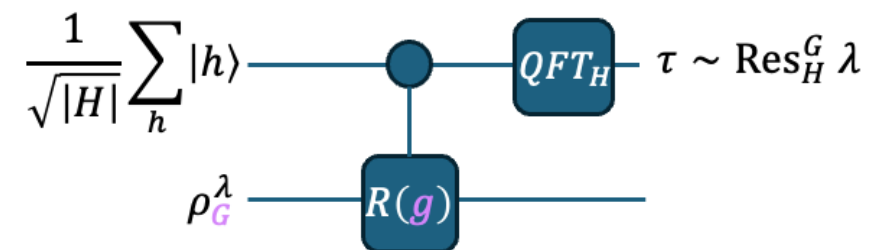
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Pseudorandom Groups

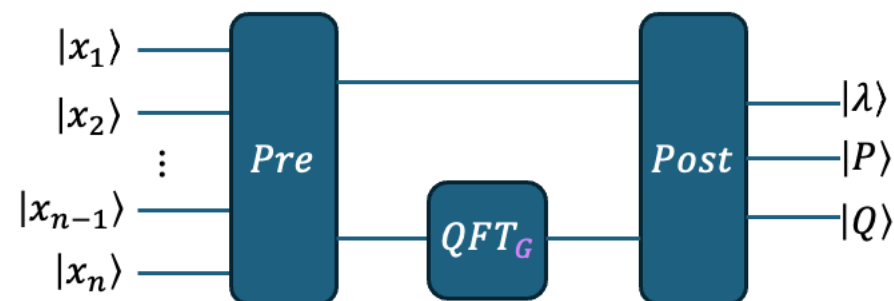
[FLMNW, Upcoming]



- Two exciting directions in quantum algorithms.



[BCGHZ 23, LH 24, CHPPW 26]



[Krovi 19, BFGLOTV 25]

Applications

Hidden Subalgebras

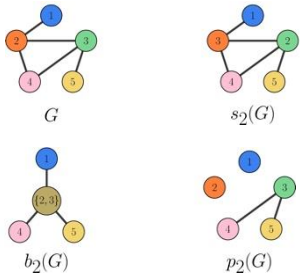
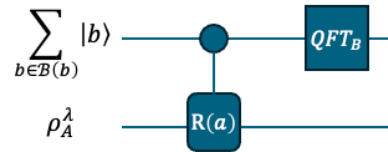


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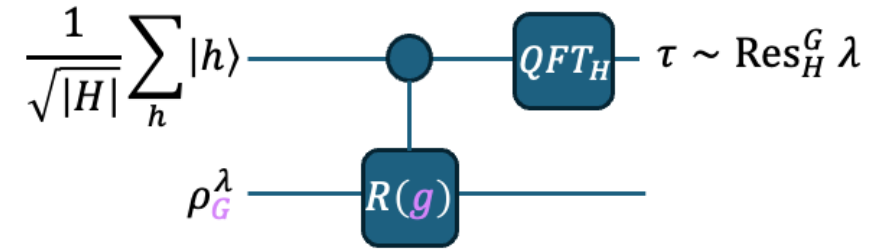
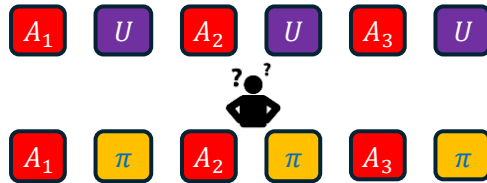
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Pseudorandom Groups

[FLMNW, Upcoming]



[BCGHZ 23, LH 24, CHPPW 26]

- How does a **group** symmetry decompose as subgroup symmetries?
- Hard to compute classically [IMW 17].
- [Panova 26]: For $G = S_n$, [LH 24] can be partially dequantized.

Applications

Hidden Subalgebras

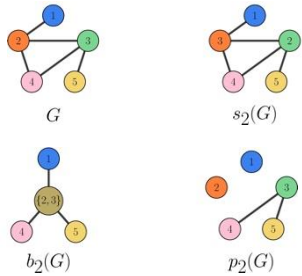
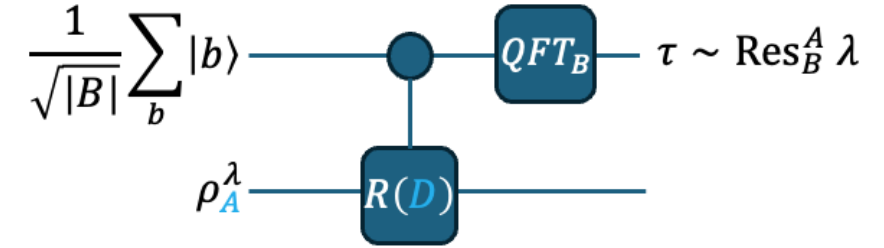
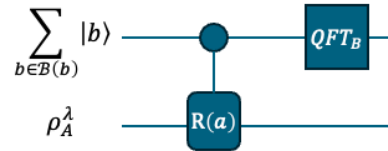


Diagram Irrep Multiplicities [LH 24, CHPPW 26]



Generalized Schur Transforms

[BCH 05, Krovı 19]

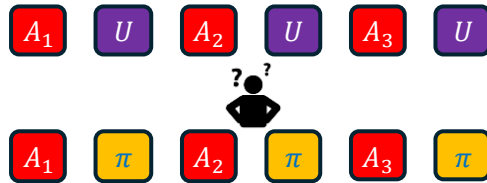
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Pseudorandom Groups

[FLMNW, Upcoming]



[BCGHZ 23, LH 24, CHPPW 26]

- How does an algebra symmetry decompose as subalgebra symmetries?
- Can these be dequantized?

Applications

Hidden Subalgebras

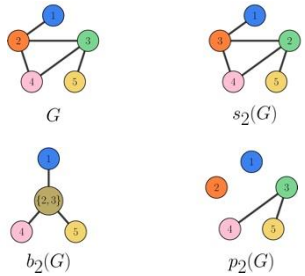
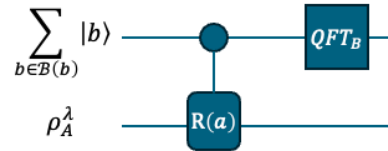


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Generalized Schur Transforms

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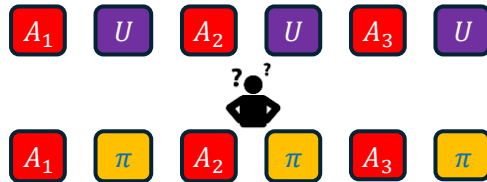
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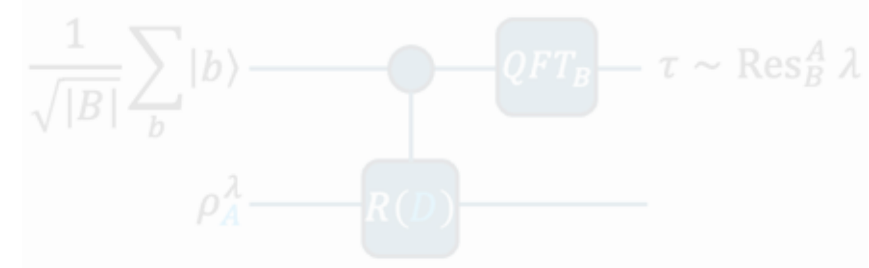
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Pseudorandom Groups

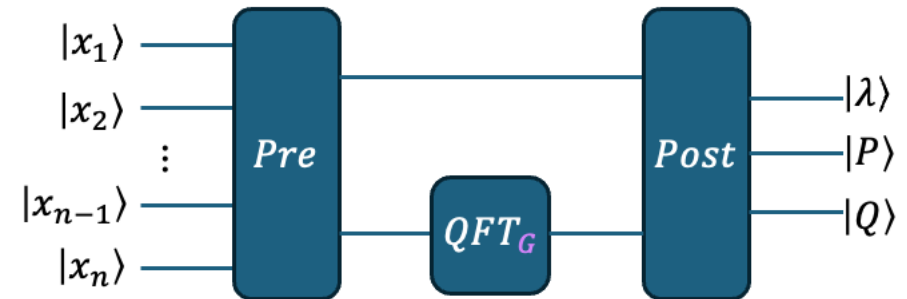
[FLMNW, Upcoming]



- The Schur Transform has wide applications [KW 01, OW 15, YCH 16, GO 22, GBO 24, GY 25, PSTW 25, FLMNW 26].



[BCGHZ 23, LH 24, CHPPW 26]



[Krovı 19, BFGLOTV 25]

Applications

Hidden Subalgebras

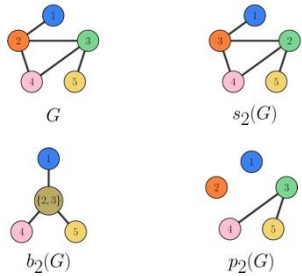
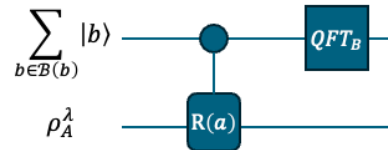


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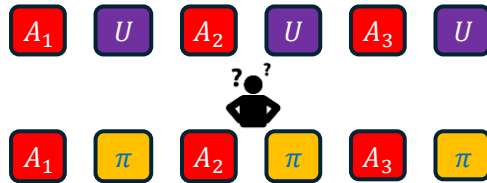
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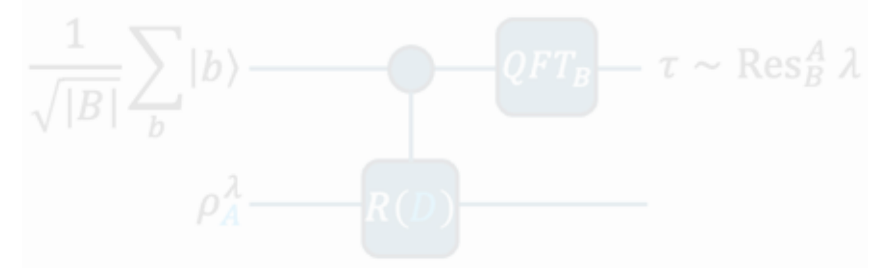
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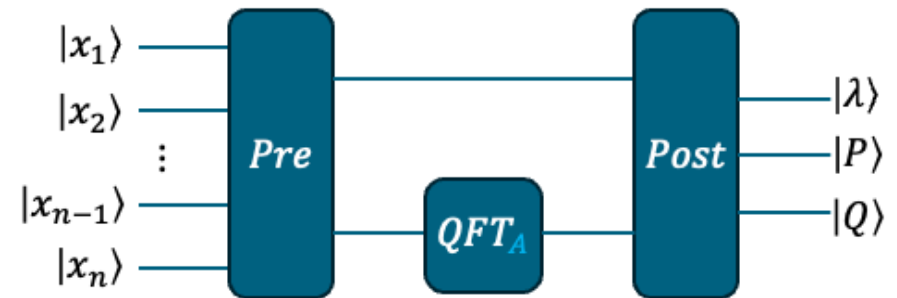
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Applications

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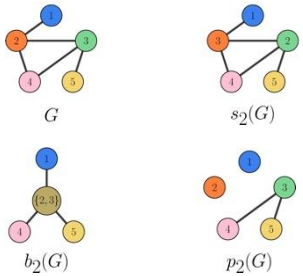
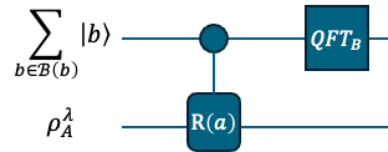


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Generalized Schur Transforms

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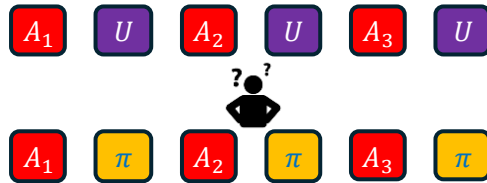
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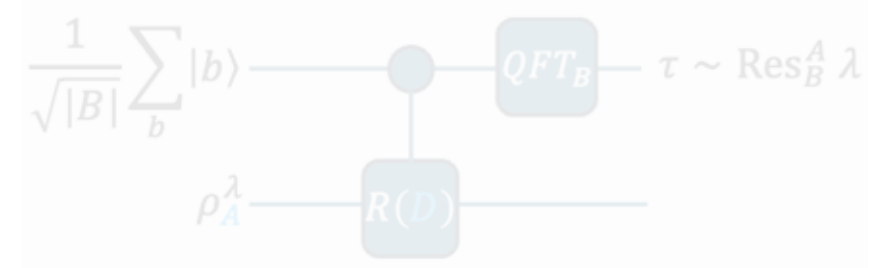
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Pseudorandom Groups

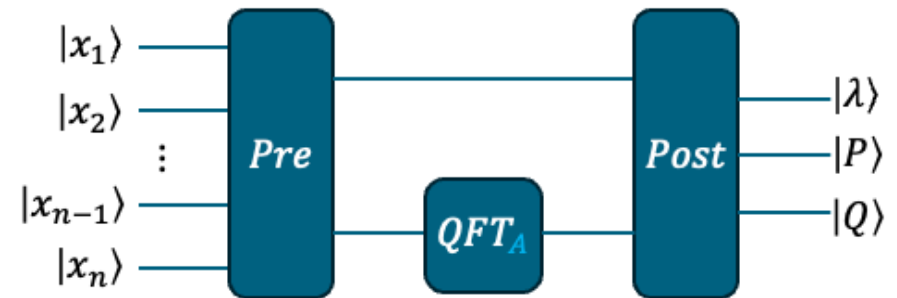
[FLMNW, Upcoming]



- The **Generalized** Schur Transform has wide applications [KW 01, OW 15, YCH 16, **GO 22**, **GBO 24**, GY 25, PSTW 25, **FLMNW 26**].
- In [FLMNW], we use it (together with techniques from this work) to give new constructions of PRUs.



[BCGHZ 23, LH 24, CHPPW 26]



[Krovı 19, BFGLOTV 25]

Implementing the Diagram Algebra QFT

QFT_A is $\exp(n \log n) / \sqrt{d}$ - close to unitary.

$$U_D = QFT_A \begin{pmatrix} \boxed{\lambda_1(D)} & & & & \\ & \boxed{\lambda_1(D)} & & & \\ & & \boxed{\lambda_2(D)} & & \\ & & & \boxed{\lambda_2(D)} & \\ 0 & & & & \boxed{\lambda_3(D)} \\ & & & & & \boxed{\lambda_4(D)} \end{pmatrix} QFT_A^\dagger$$

Implementing the Diagram Algebra QFT

QFT_A is $\exp(n \log n) / \sqrt{d}$ - close to unitary, and is approximated by a $\text{poly}(n, \log d)$ -size circuit.

$$U_D = QFT_A \begin{pmatrix} \lambda_1(D) & & & & \\ & \lambda_1(D) & & & \\ & & \lambda_2(D) & & \\ & & & \lambda_2(D) & \\ 0 & & & & \lambda_3(D) \\ & & & & & \lambda_4(D) \end{pmatrix} QFT_A^\dagger$$

Efficient Fourier Transforms

	Classical Algorithms	Quantum Algorithms
Group (Algebras)		
Non-Group (Algebras)		

Efficient Fourier Transforms

	Classical Algorithms	Quantum Algorithms
Group (Algebras)	Abelian Groups [CT 65] Non-Abelian Groups [Beth 87, DR 90]	
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Efficient Fourier Transforms

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Efficient Fourier Transforms

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Non-Group (Algebras)	Diagram Algebras [MRV 16] Steps Generalize	Diagram Algebras* Generalizes, but many new ideas required.

The recurring approach: recursion (**Separation of Variables**)

Reduce the QFT on a supergroup (superalgebra) to the QFT on a subgroup (subalgebra).

Separation of Variables

For Group Algebras

On input $|\psi_{in}\rangle = \sum_g f(g)|g\rangle$:

For General Algebras

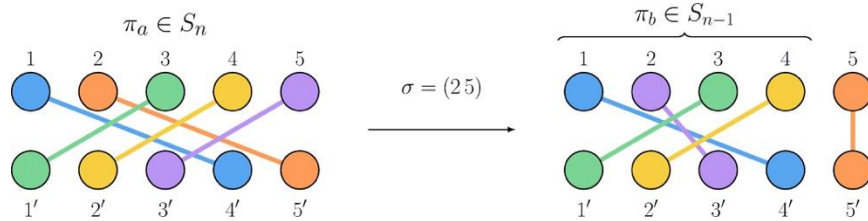
On input $|\psi_{in}\rangle = \sum_{a \in \mathcal{B}(A)} f(a)|a\rangle$:

Separation of Variables

For $\mathbb{C}[S_n]$:

On input $|\psi_{in}\rangle = \sum_{\pi} f(\pi) |\pi\rangle$:

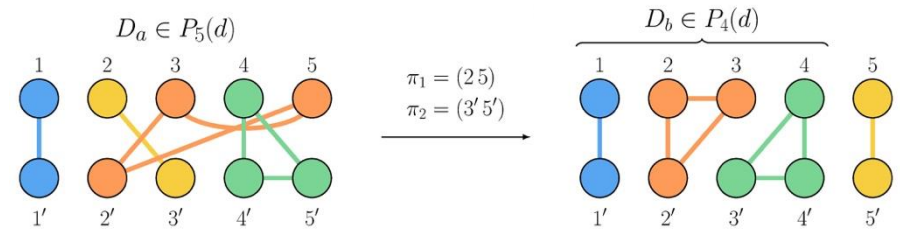
Factor the input into a subgroup
(subalgebra), and a “coset” representative



Since S_n is a group, there is a unique (left) coset.

For a Diagram Algebra:

On input $|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle$:



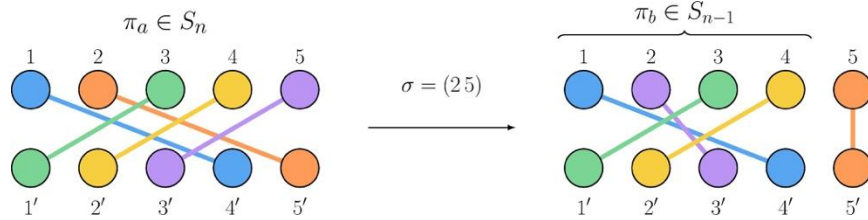
Need a left and right “coset”, and the factorization *is no longer unique*.

Separation of Variables

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Factor the input into a subgroup
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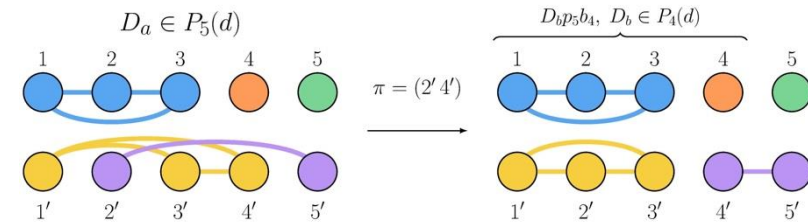


Since S_n is a group, there is a unique (left) coset.

$$\sum_w \sum_{\sigma \in S_{n-1}} f(w\sigma) |w\rangle \otimes |\sigma\rangle$$

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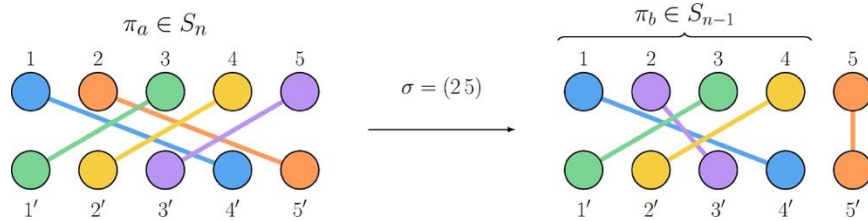
Also, we can't always perfectly factor.

Separation of Variables

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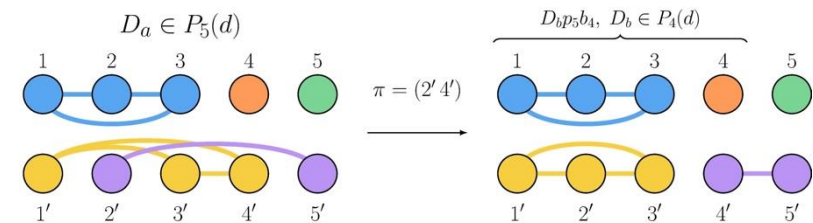


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Also, we can't always perfectly factor.

$$\sum_{w_1, w_2} \sum_{E \in \mathcal{C}(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes |E\rangle \otimes |w_2\rangle$$

Separation of Variables

For $\mathbb{C}[S_n]$:

On input $|\psi_{in}\rangle = \sum_{\pi} f(\pi) |\pi\rangle$:

Factor the input into a subgroup
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$$\sum_w \sum_{\sigma \in S_{n-1}} f(w\sigma) |w\rangle \otimes |\sigma\rangle$$

Recursively apply the Fourier Transform

$$\sum_w \sum_{\sigma \in S_{n-1}} f(w\sigma) |w\rangle \otimes QFT_{S_{n-1}} |\sigma\rangle$$

For a Diagram Algebra:

On input $|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle$:

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes |E\rangle \otimes |w_2\rangle$$

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_B |E\rangle \otimes |w_2\rangle$$

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$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_B |E\rangle \otimes |w_2\rangle$$

Embed this Fourier state into the larger algebra

$$\sum_w \sum_{\sigma \in S_{n-1}} f(w\sigma) |w\rangle \otimes QFT_{S_n} |\sigma\rangle$$

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |E\rangle \otimes |w_2\rangle$$

Separation of Variables

For $\mathbb{C}[S_n]$:

On input $|\psi_{in}\rangle = \sum_{\pi} f(\pi) |\pi\rangle$:

Embed this Fourier state into the larger algebra

$$\sum_w \sum_{\sigma \in S_{n-1}} f(w\sigma) |w\rangle \otimes QFT_{S_n} |\sigma\rangle$$



Superposition of $|\lambda, i, j\rangle$

For a Diagram Algebra:

On input $|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle$:

$$\sum_{w_1, w_2} \sum_{E \in \mathcal{C}(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |E\rangle \otimes |w_2\rangle$$



Superposition of $|\lambda, i, j\rangle$

- Achieved by rotating $|\lambda, i, j\rangle$ to $|\lambda', i \circ \lambda', j \circ \lambda'\rangle$ with angle $\|E_{i \circ \lambda', j \circ \lambda'}^{\lambda'}\|_2 / \|E_{ij}^{\lambda}\|_2$.
- $\|E_{ij}^{\lambda}\|_2^2 \propto$ dimension of the irrep in the commutant, which can be efficiently computed.

$$[U(d)^{\otimes n}, \mathbb{C}[S_n]] = 0$$

$$[O(d)^{\otimes n}, B_n(d)] = 0$$

$$[\pi^{\otimes n}, P_n(d)] = 0$$

Separation of Variables

For $\mathbb{C}[S_n]$:

On input $|\psi_{in}\rangle = \sum_{\pi} f(\pi) |\pi\rangle \dots$

$$\sum_w \sum_{\sigma \in S_{n-1}} f(w\sigma) |w\rangle \otimes QFT_{S_n} |\sigma\rangle$$



Superposition of $|\lambda', i \circ \lambda', j \circ \lambda'\rangle$

For a Diagram Algebra:

On input $|\psi_{in}\rangle = \sum_{D \in B(A)} f(D) |D\rangle \dots$

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |E\rangle \otimes |w_2\rangle$$



Superposition of $|\lambda', i \circ \lambda', j \circ \lambda'\rangle$

- Next, controlled on $|w\rangle$ and $|\lambda'\rangle$, apply the irrep matrix $\lambda'(w)$.

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |w_1 E w_2\rangle \otimes |w_2\rangle$$

$$U_w = QFT_A \begin{pmatrix} \lambda_1(w) & & & & \\ & \lambda_1(w) & & & \\ & & \lambda_2(w) & & \\ & & & \lambda_2(w) & \\ 0 & & & & \lambda_3(w) \\ & & & & & \lambda_4(w) \end{pmatrix} QFT_A^\dagger$$

Separation of Variables

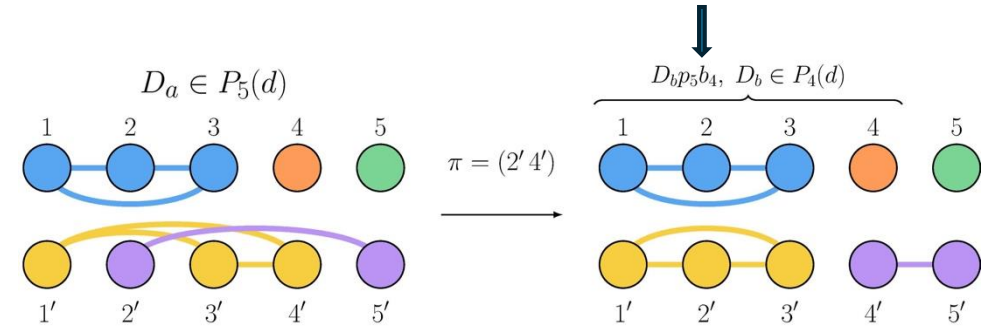
For a Diagram Algebra:

On input $|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle \dots$

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |E\rangle \otimes |w_2\rangle$$

Superposition of $|\lambda', i \circ \lambda', j \circ \lambda'\rangle$

If w is not a permutation, then $\lambda'(w)$ is not unitary!



We carefully handle these cases by combining them with the embedding step.

- Next, controlled on $|w\rangle$ and $|\lambda'\rangle$, apply the irrep matrix $\lambda'(w)$.

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |w_1 E w_2\rangle \otimes |w_2\rangle$$

$$U_w = QFT_A \begin{pmatrix} \lambda_1(w) & & & & \\ & \lambda_1(w) & & & \\ & & \lambda_2(w) & & \\ & & & \lambda_2(w) & \\ 0 & & & & \lambda_3(w) \\ & & & & & \lambda_4(w) \end{pmatrix} QFT_A^\dagger$$

Separation of Variables

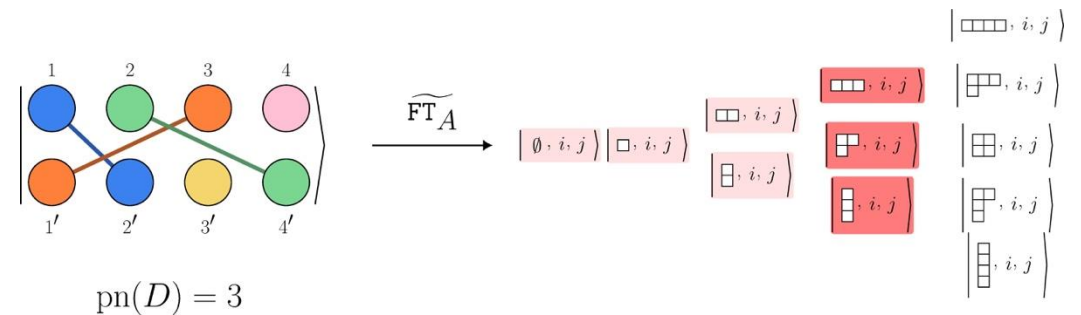
For a Diagram Algebra:

On input $|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle \dots$

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |E\rangle \otimes |w_2\rangle$$

Superposition of $|\lambda', i \circ \lambda', j \circ \lambda'\rangle$

If w is not a permutation, then $\lambda'(w)$ is not unitary!



Crucial to the analysis is that the diagram algebra QFT transforms *propagating number* to *box count*.

- Next, controlled on $|w\rangle$ and $|\lambda'\rangle$, apply the irrep matrix $\lambda'(w)$.

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |w_1 E w_2\rangle \otimes |w_2\rangle$$

$$U_w = QFT_A \begin{pmatrix} \lambda_1(w) & & & & \\ & \lambda_2(w) & & & \\ & & 0 & & \\ & & & \lambda_3(w) & \\ 0 & & & & \lambda_4(w) \end{pmatrix} QFT_A^\dagger$$

Separation of Variables

On input

$$|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle$$

We now have the state

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) |w_1\rangle \otimes QFT_A |w_1 E w_2\rangle \otimes |w_2\rangle$$

$(|w_1 E w_2\rangle = |D\rangle)$

- If we could only guess $|w_1\rangle$ and $|w_2\rangle$, we could uncompute them.
- For **groups**, there is one correct guess, so we “try every possible coset” [Beals 97].
- For **algebras**, there are many “correct” guesses (non-unique factorization).
 - The second “correct” guess will undo the first!

Strategy: Guess $(|w_1\rangle, |w_2\rangle)$ in an order such that the algorithm only ever detects the “*last possible*” factorization.

Separation of Variables

On input

$$|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle$$

We now have the state

$$\sum_{w_1, w_2} \sum_{E \in C(w_1, w_2)} f(w_1 E w_2) QFT_A |w_1 E w_2\rangle$$

$$(|w_1 E w_2\rangle = |D\rangle)$$

Separation of Variables

On input

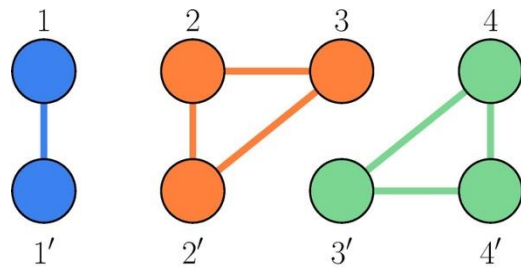
$$|\psi_{in}\rangle = \sum_{D \in \mathcal{B}(A)} f(D) |D\rangle$$

We now have the state

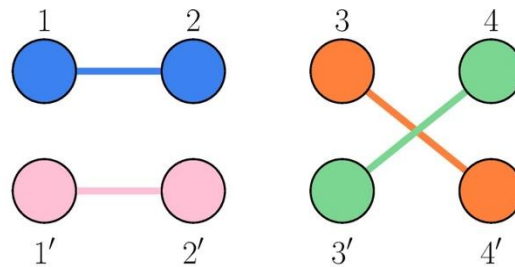
$$\sum_{D \in \mathcal{B}(A)} f(D) QFT_A |D\rangle$$



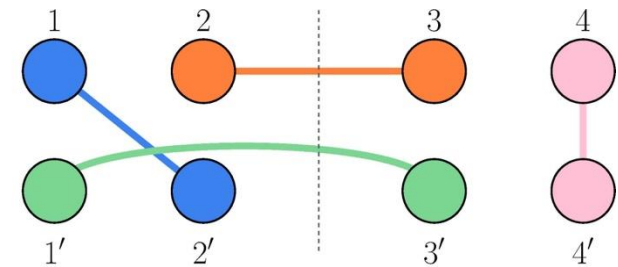
QFT_A is $\exp(n \log n) / \sqrt{d}$ - close to unitary, and is approximated by a $\text{poly}(n, \log d)$ -size circuit.



$D \in P_4(d)$



$D \in B_4(d)$



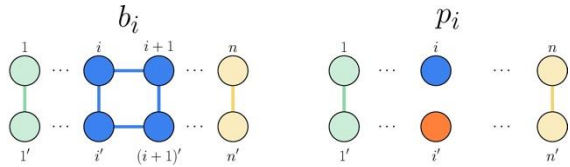
$D \in B_{2,2}(d)$

Thank you!

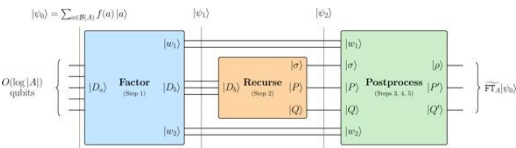


The Fourier Transform and Diagram Algebras

$$\text{FT}_G U_g \text{FT}_G^\dagger = \begin{pmatrix} \rho_1(g) & & & \\ & \rho_2(g) & & \\ & & \rho_2(g) & \\ & & & \ddots \\ & & & & \rho_n(g) \end{pmatrix}$$

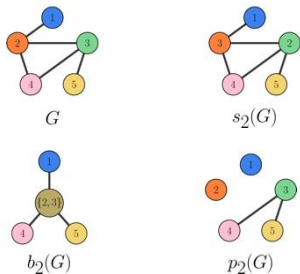


Upgrading the “Separation of Variables Approach” [DR 90, MRR 03, MRW 16]



- Non-Unique Factorization
- “Generalized” Embeddings
- Replacing Dual Elements
- and more...

Hidden Subalgebras



Generalized Schur Transforms

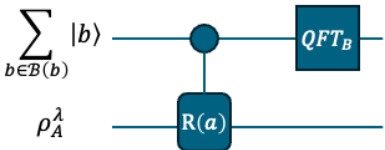
[BCH 05, Krovi 19]

$$[U(d)^{\otimes n}, \mathbb{C}[S_n]] = 0$$

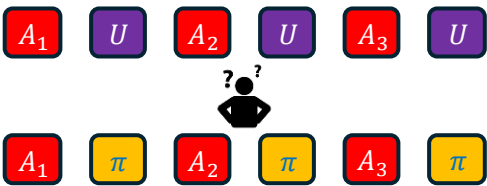
$$[O(d)^{\otimes n}, B_n(d)] = 0$$

$$[\pi^{\otimes n}, P_n(d)] = 0$$

Diagram Irrep Multiplicities [LH 24, CHPPW 26]

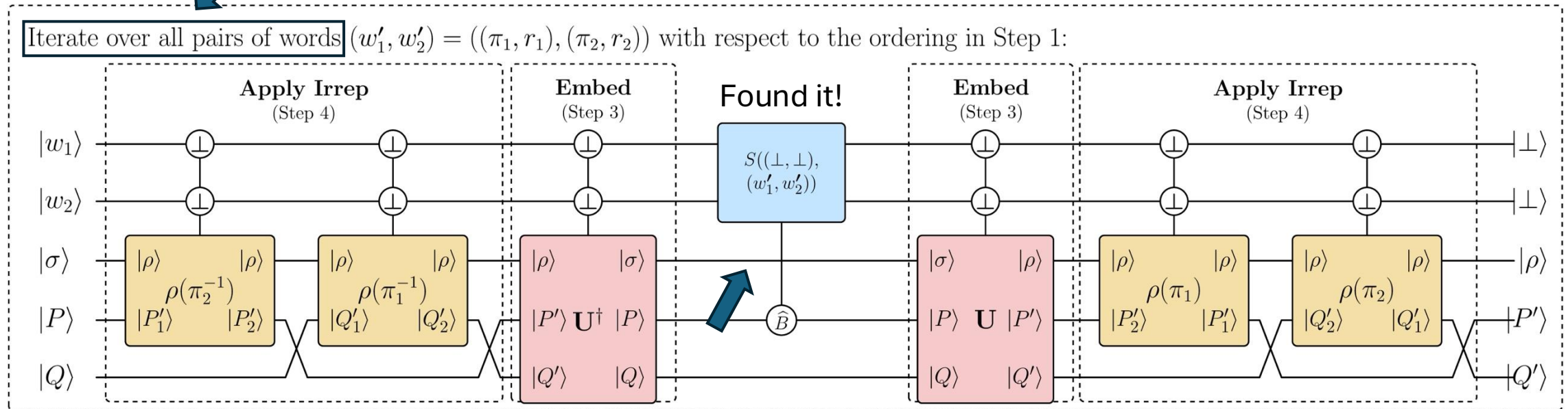


Pseudorandom Groups [FLMNW, Upcoming]



Extra Slide: Summing over Cosets

“Guess” the Transversal



For future guesses, do nothing
(relies on the last possible factorization) .

Fix up the recursion.